//MATLAB CODE

```
mp = 0.2;

lp = 0.3;

la = 0.2;

Jp = 0.006;

Ja = 0.002;

g = 9.81;

b_theta = 0.01;

b_phi   = 0.02;

A1 = Ja + mp*la^2;

B1 = mp*la*lp;

C1 = Jp + mp*lp^2;

M = [A1 B1;

   B1 C1];

M_inv = inv(M);

A = [0 1 0 0;

   0 0  M_inv(1,2)*mp*g*lp  0;

   0 0 0 1;

   0 0  M_inv(2,2)*mp*g*lp  0]

B = [0;

   M_inv(1,1);

   0;

   M_inv(2,1)]

C = [0 0 1 0];        % pendulum angle output
```

```
D = 0;

sys = ss(A,B,C,D);

disp('Open loop eigenvalues:')

eig(A)

disp('Transfer Function of the System:')

G = tf(sys)

t = 0:0.01:10;

x0 = [0.3 0 0.1 0];

[~,t,x] = initial(sys,x0,t);

figure

plot(t,x(:,3),'LineWidth',2)

xlabel('Time (seconds)')

ylabel('\phi (Pendulum Angle)')

title('Open Loop Response (Unstable)')

grid on

Q = diag([10 1 100 1]);

R = 1;

K = lqr(A,B,Q,R);

disp('LQR Gain Matrix:')

K

Acl = A - B*K

disp('Closed loop eigenvalues:')

eig(Acl)

sys_cl = ss(Acl,B,C,D);
```

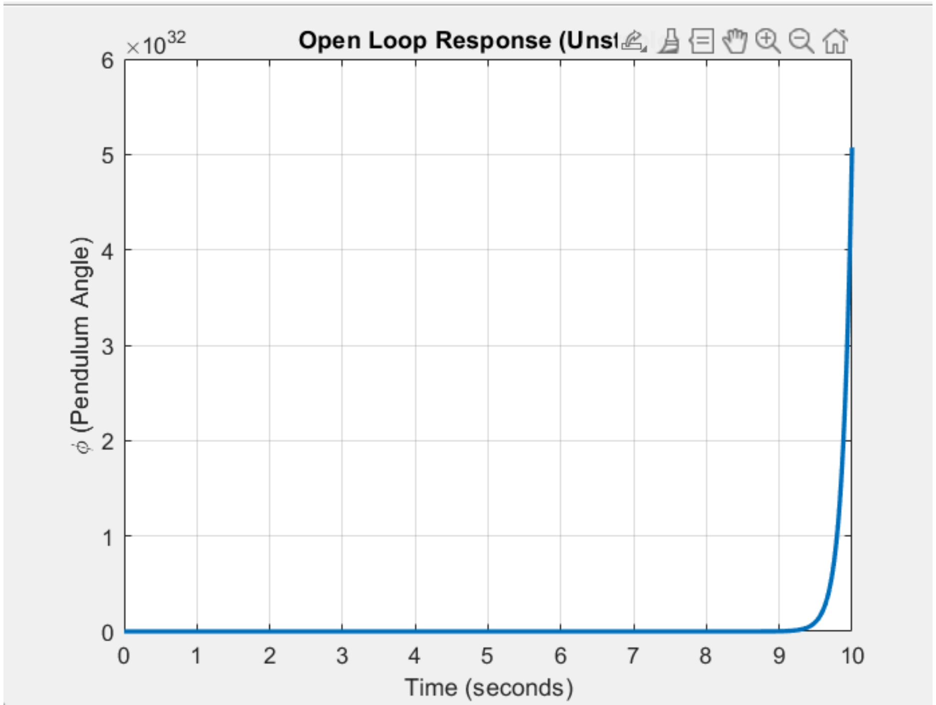
```
disp('Transfer Function of the System:')

G_cl = tf(sys_cl)

t = 0:0.01:5;

x0 = [0.3 0 0.1 0];

[~,t,x] = initial(sys_cl,x0,t);

figure

plot(t,x(:,3),'LineWidth',2)

title('Closed Loop Response (LQR Stabilized)')

xlabel('Time')

ylabel('\phi (Pendulum Angle)')

grid on

t = 0:0.01:5;

[y,t,x] = initial(sys_cl,[0.3 0 0.1 0],t);

figure

plot(t,x(:,1),'LineWidth',2)

hold on

plot(t,x(:,2),'LineWidth',2)

plot(t,x(:,3),'LineWidth',2)

plot(t,x(:,4),'LineWidth',2)

xlabel('Time')

ylabel('States')

title('Closed Loop State Response')

legend('\theta','\phi','\theta dot','\phi dot')

grid on
```

RESULTS :

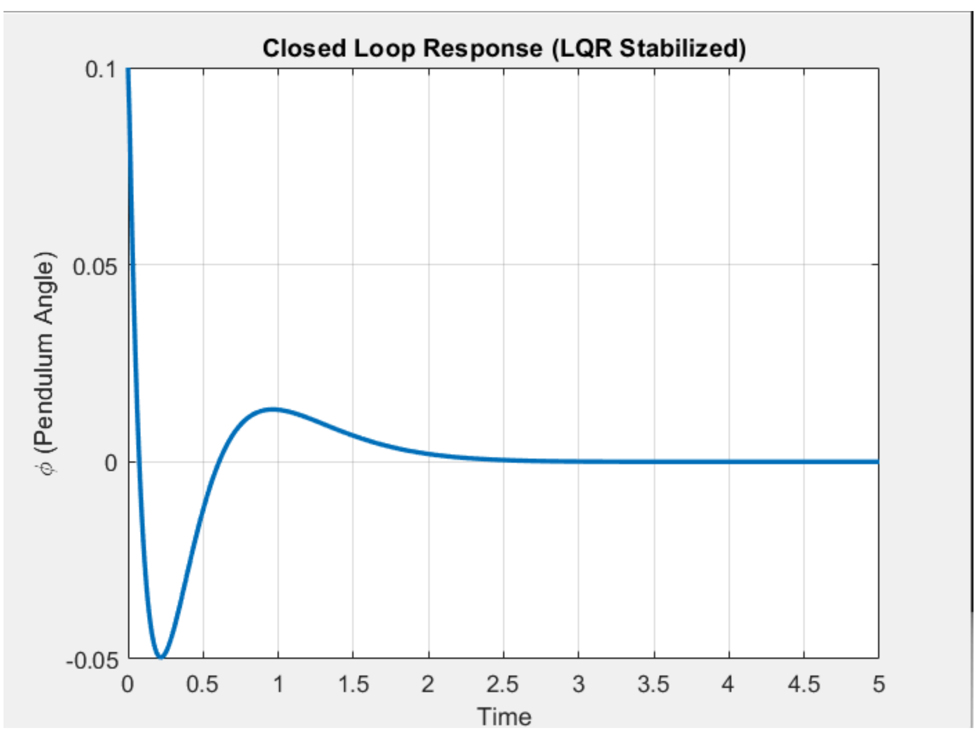
```
>> A1_24EC10125_Controls

A =

         0    1.0000         0         0
         0         0  -73.5750         0
         0         0         0    1.0000
         0         0   61.3125         0


B =

     0
   250
     0
  -125

Open loop eigenvalues:

ans =

         0
         0
    7.8302
   -7.8302

Transfer Function of the System:

G =

            -125
  -------------------------
  s^2 - 8.882e-16 s - 61.31

Continuous-time transfer function.
Model Properties
LQR Gain Matrix:

K =

   -3.1623   -2.3747  -36.4700   -7.0909
```

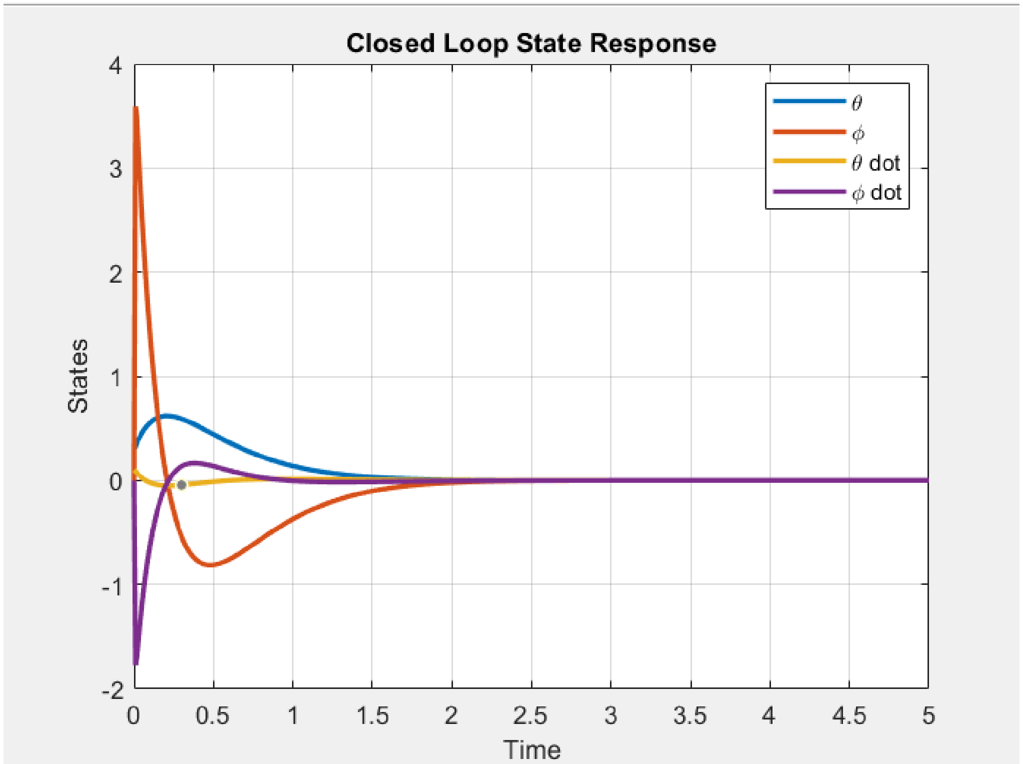
```
Acl =

   1.0e+03 *

         0    0.0010         0         0
    0.7906    0.5937    9.0439    1.7727
         0         0         0    0.0010
   -0.3953   -0.2968   -4.4974   -0.8864

Closed loop eigenvalues:

ans =

   1.0e+02 *

  -2.7961 + 0.0000i
  -0.0723 + 0.0000i
  -0.0292 + 0.0103i
  -0.0292 - 0.0103i

Transfer Function of the System:

G_cl =

          -125 s^2 - 5.928e-12 s - 1.321e-11
  ---------------------------------------------------
```

s^4 + 292.7 s^3 + 3707 s^2 + 1.456e04 s + 1.93



9e04




Closed Loop State Response
States
Time
θ
ϕ
θ dot
ϕ dot
4
3
2
1
0
-1
-2
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5